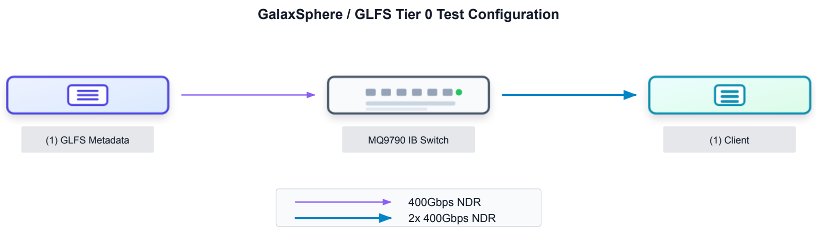
GalaxSphere-1Tier0

System Description

System Name	GalaxSphere-1Tier0
Submitter Name	GalaxSphere
Submission Date	July 24, 2026
Benchmark Version	3.0
Solution Type	Distributed parallel file system with localization capability (Tier0)
Submission Division	CLOSED
Workloads	3D U-Net B200、RetinaNet B200、Checkpointing Llama 3 8B、Checkpointing Llama 3 70B、KVCache、VectorDB

System Topology



For these test runs, GalaxSphere is deployed with a dedicated metadata node, an InfiniBand fabric, and a single benchmark client. The metadata node maintains the global namespace, tracks file and chunk metadata, coordinates cluster state, and decides how file data is placed across available storage resources.

The benchmark client serves two roles. It runs the MLPerf Storage workload and also contributes local NVMe devices to the GalaxSphere storage pool as chunk-server capacity. In this Tier 0 configuration, those client-side NVMe devices are treated as shared storage resources managed by GalaxSphere, while the metadata service controls file layout and distribution.

The client accesses the system through a POSIX-compatible GalaxSphere file-system mount. Applications issue normal file I/O through this mount point. Metadata requests are handled by the metadata service, while bulk data reads and writes are served directly from the chunk-server path backed by the NVMe devices, keeping data movement out of the metadata node.

Hardware

The hardware configuration used for these runs is detailed in the tables below.

Metadata Node

Resource	Qty	Specification
System	1	H3C UniServer R4700, Ubuntu 24.04.3 LTS, kernel 6.8.0-71-generic
CPU	2	Intel Xeon Platinum 8473C, 52 cores/socket, 104 total cores, 1 thread/core, 2.1 GHz
Memory	4	4 x 16GB Hynix DDR5 RDIMM, 64GB
Boot Drive	1	KINGSTON SNV3S500G NVMe, 500.11GB / 465.8GiB
Network Adapters	1	Mellanox ConnectX-7 InfiniBand, MT2910 Family
Storage Drives	1	KINGSTON SNV3S500G NVMe, 500.11GB / 465.8GiB

MLPerf Client

Resource	Qty	Specification
System	1	AIC HA2026-HC, BIOS Hera_1.03.0, Ubuntu 24.04.3 LTS, kernel 6.8.0-71-generic
CPU	1	AMD EPYC 9655, 96 cores/socket, 192 total threads, 2 threads/core, 2.6 GHz
Memory	12	12 x 32GB Micron DDR5 RDIMM, 384GB
Boot Drive	1	KINGSTON SNV3S500G NVMe, 500.11GB / 465.8GiB
Network Adapters	2	Mellanox ConnectX-7 InfiniBand, MT2910 Family
Storage Drives	24	16 x ScaleFlux CSDU7UXF38 NVMe 1.92TB + 8 x ScaleFlux CSDU7UVG76 NVMe 3.84TB

Network

Resource	Qty	Specification
InfiniBand Switch	1	QM9790 1U NDR 400Gbps InfiniBand Switch

Storage Rack Units

Resource	Qty	Rack U each	Total Rack Units
Metadata Node	1	1	1
Configuration Total			1

Power Requirements

System Component	PS Unit	Nameplate Rated Power	Design Power
Metadata Node	Power supply 1	1600 W	1600 W
	Power supply 2	1600 W	
MLPerf Client #1	Power supply 1	3200 W	3200 W
	Power supply 2	3200 W	
InfiniBand Switch	Power supply 1	1600 W	1600 W
	Power supply 2	1600 W	
Configuration Total	/		6400 W

Software

The software running on each component type in this configuration is detailed below.

Metadata

GalaxSphere / GLFS is deployed as a Linux-based metadata service with GLFS management and metadata components installed on the metadata node. The metadata node provides namespace management, file metadata operations, layout decisions, and cluster coordination for the storage system.

- GalaxSphere / GLFS metadata service, version 1.0.7
- Operating system: Ubuntu 24.04.3 LTS
- Kernel: 6.8.0-71-generic
- Metadata node hardware: H3C UniServer R4700

MLPerf Client

The MLPerf client runs the standard, unmodified MLPerf Storage benchmark code. In this configuration, the client also runs the GLFS client stack and acts as a chunk server, contributing its local NVMe devices as Tier 0 storage resources managed by GLFS.

- Standard, unmodified MLPerf Storage benchmark code
- Operating system: Ubuntu 24.04.3 LTS
- Kernel: 6.8.0-71-generic
- GalaxSphere / GLFS client stack and chunk server, version 1.0.7
- GLFS mounted through a POSIX-compatible private file system
- Client-local NVMe devices used as GLFS Tier 0 chunk storage

Network Switch

The metadata node and MLPerf client are connected through an InfiniBand fabric. The network is used for metadata communication, cluster coordination, and data I/O between the GLFS client/chunk-server components.

- InfiniBand switch fabric
- Client adapter: Mellanox ConnectX-7 InfiniBand
- Metadata/control and data traffic carried over the test fabric

Settings

Metadata Node

No non-default tunings were applied.

MLPerf Client

No non-default tunings were applied.

Network Switch

No non-default tunings were applied.